

Review of solid–liquid phase change materials and their encapsulation technologies

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ABSTRACT

Various types of solid–liquid phase change materials (PCMs) have been reviewed for thermal energy storage applications. The review has shown that organic solid–liquid PCMs have much more advantages and capabilities than inorganic PCMs but do possess low thermal conductivity and density as well as being flammable. Inorganic PCMs possess higher heat storage capacities and conductivities, cheaper and readily available as well as being non-flammable, but do experience supercooling and phase segregation problems during phase change process. The review has also shown that eutectic PCMs have unique advantage since their melting points can be adjusted. In addition, they have relatively high thermal conductivity and density but they possess low latent and specific heat capacities. Encapsulation technologies and shell materials have also been examined and limitations established. The morphology of particles was identified as a key influencing factor on the thermal and chemical stability and the mechanical strength of encapsulated PCMs. In general, in-situ polymerization method appears to offer the best technological approach in terms of encapsulation efficiency and structural integrity of core material. There is however the need for the development of enhancement methods and standardization of testing procedures for microencapsulated PCMs.

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List of symbols

(OP)-10	polyethylene glycol octylphenyl ether	PETRA	pentaerythritol tetraacrylate
ABS	acrylonitrile–styrene–butadiene copolymer	PFR	phenolic resin
AS	acrylonitrile–styrene copolymer	PMMA	polymethylmethacrylate
BDDA	1,4-butyleneglycol diacrylate	PS	polystyrene
CAB	cellulose acetate butyrate	PSB	styrene-1,4-butylene glycol diacrylate copolymer
Cp	specific heat	PSD	styrene–divinylbenzene copolymer
DETA	diethylene triamine	PSDB	styreneedivinylbenzene 1,4-butylene glycol diacrylate copolymer
DNS-86	ether sulfate	PVAc	polyvinyl acetate
DSC	differential scanning calorimetry	SA	stearic acid
DSP	sodium phosphate dodecahydrate	SDS	sodium dodecyl sulphate
DVB	divinylbenzene	SEM	scanning electron microscope
DVB	divinyl benzene	SEM	scanning electron microscopy
EDA	ethylene diamine	SMA	styrene–maleic anhydride-monomethyl
EMT	effective medium theory	St	styrene
FT-IR	Fourier transform infrared	St-BA	styrene-butyl acrylate
<i>H</i>	latent heat	St-MMA	styrene-methyl methacrylate
HD	hexadecane	TA	sodium laureth sulfate
HSMA	hydrolyzed-styrene-alt-maleic anhydride	TDI	tolylene 2,4-diisocyanate
<i>k</i>	thermal conductivity	TEM	transmission electron microscope
MDI	methylene diisocyanate	TG	thermogravimetric
MEPCM	microencapsulated phase change material	TGA	thermal gravimetric analysis
MF	melamine formaldehyde	<i>T_m</i>	melting temperature
n.v.n.s.	non-volatile non-solvent	TMPTA	trimethylol propane triacrylate
NEPCM	nanoencapsulated phase change material	TPGDA	tripropylene glycol diacrylate
O/W	oil/water	TSC	two-step coacervation
PA	palmitic acid	UF	urea formaldehyde
PC	polycarbonate	v.s.	volatile solvent
PCMs	phase change materials	WBPU	waterborne polyurethane
PDVB	polydivinylbenzene	XRD	X-ray diffraction
PEMA	polyethyl methacrylate	ρ	density

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1. Introduction

Energy consumption in buildings continues to pose environmental problems to many countries and the world as a whole. Techniques such as thermal energy storage are being explored at different levels for reducing energy consumption in buildings which currently accounts for about 40% of total global energy consumption [1]. Phase change materials (PCMs) are capable of storing and releasing large amounts of energy during melting and solidification at specific temperatures. Thermal energy storage does not only reduce the mismatch between energy supply and demand but also improves the performance and reliability of energy systems and plays an important role in conserving energy resources. Current applications of PCMs in buildings include air conditioning, i.e., free cooling [1], cold thermal storage media and absorption refrigeration. Other integrated systems are PCM Trombe wall, PCM wallboards, PCM shutter, PCM concrete, PCM

under-floor heating systems, PCM ceiling boards [2–4] as well as hot water supply and waste heat recovery systems [5]. For instance, Oro et al. [6] and Li et al. [7] reviewed PCMs melting point below 20 °C for cold thermal energy storage applications. Agyenim et al. [8] identified phase change materials of melting temperature within 0–65 °C to be suitable for domestic heating/cooling application. They also stated that PCMs of melting temperatures 80–120 °C could be used in absorption cooling system, whereas those types of melting temperatures above 150 °C could be applied in solar power plants systems coupled with parabolic trough collectors for direct steam generation. Furthermore, Cabeza et al. [9] stated more comprehensively that melting temperatures up to 21 °C are more suitable for cooling applications, 22–28 °C for thermal comfort applications, 29–60 °C for hot water supply and over 120 °C for waste heat recovery applications.

Depending on the type of PCM, energy storage process could be described as solid–solid, solid–liquid, liquid–gas or solid–gas as

shown in Fig. 1 [2,3,8,9]. However, liquid–gas and solid–gas processes are not applicable to construction materials due to their large volume and pressure change during phase change process. Solid–solid PCMs change their crystalline structure from one lattice configuration to another at a fixed temperature without supercooling, visible change and leakage but cannot easily be mixed with other construction materials (i.e. cement, gypsum). On the other hand, solid–liquid types of organic PCMs could leak into their surroundings during heat storage process if they are applied directly without being encapsulated. Most of the organic PCMs also have low thermal conductivity and poor thermal response as well as being flammable, which pose a serious potential danger of fire outbreak. Meanwhile, most inorganic PCMs are known to be corrosive, which could cause irretrievable damage to storage containers [10]. These problems may be overcome by employing encapsulation technology to produce enhanced microencapsulated phase change material (MEPCM) or nanoencapsulated phase change material (NEPCM) for various applications. As shown in Fig. 2, a typical structure of MEPCM/NEPCM is composed of core and shell layers [11], which can be classified as mononuclear, polynuclear or matrix type. This paper therefore focusses on review of various types of solid–liquid PCMs and their related micro-/nano-encapsulation technologies in order to advance their development and application.

2. Properties of solid–liquid PCMs

There are a number of factors which can influence effectiveness and application of solid–liquid types of PCMs. For instance, good thermal properties depend on suitable phase transition temperature, high latent heat of transition per unit weight, high thermal conductivity, and large specific heat capacity. In the physical side PCMs should have favourable phase equilibrium, high density, minor volume changes during phase transition, and low vapour pressure at the operation temperature. For kinetic properties the

PCMs need to exhibit little or no sub-cooling during freezing, sufficient crystallization rate, melting and solidification at the same temperature and phase segregation. They should also possess good chemical properties capable of completing reversible freezing/melting cycle, long-term chemical stability, compatibility with other materials, no degradation after long-term thermal cycles, non-corrosive, non-toxic, no fire hazard, and non-explosive compounds. These properties are therefore reviewed in the following section for the different types of organic and inorganic PCMs.

2.1. Organic solid–liquid PCMs

Organic solid–liquid PCMs are described as paraffin and non-paraffin materials (fatty acids, alcohols and glycols). Organic materials have stable phase change temperature (without phase segregation), consequent degradation of latent heat fusion, self-nucleation (no supercooling) and usually non-corrosiveness [12].

2.1.1. Paraffin materials

Paraffin belongs to a family of saturated hydrocarbons with similar properties and molecular formulas (C_nH_{2n+2} , as shown in Table 1) with straight hydrocarbon chains. In general, the longer the length of hydrocarbon chain (more carbon atoms in paraffin molecular formulas), the higher the melting temperature. Paraffin materials are safe, reliable, predictable, less expensive, non-corrosive and have low vapour pressure. However, they possess some undesirable properties such as low thermal conductivity, non-compatible with the plastic container, moderately flammable (i.e. $15 < q < 50$ [13], where q is the flammability index), a high volume change between the solid and liquid stages. Table 1 shows the thermophysical properties of the commercially available paraffin materials.

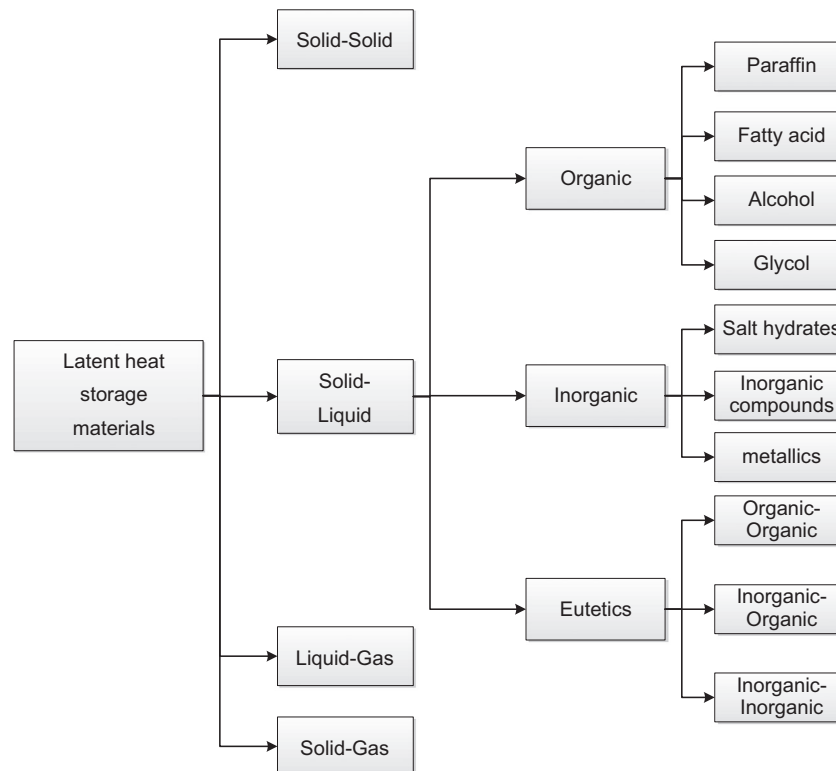


Fig. 1. Classification of PCMs.

2.1.2. Non-paraffin materials

Non-paraffin organic materials such as esters, fatty acids, alcohols and glycols, can also be used as PCMs as listed in Table 2. They are the largest group of candidate materials for latent heat storage, particularly, the fatty acids are the compounds of carbon (C), hydrate (H) and oxygen (O) with the formulas of $C_mH_nO_2$ (where m and n are the atomic numbers). They are however flammable and should therefore not be exposed to excessively high temperatures, flames or oxidizing agents. Other properties of the non-paraffin PCMs include high heat of fusion, inflammability, low thermal conductivity, low flash point, varying level of toxicity, and instability at high temperature.

2.2. Inorganic solid–liquid PCMs

2.2.1. Salt hydrates

Salt hydrates in Table 3 are the oldest and most studied PCMs, which can be regarded as alloys of inorganic salts and water forming a typical crystalline solid of general formula $A \cdot nH_2O$ (where A is the salt component, n is the molecular number and H_2O represents water). The solid–liquid transformation of salt hydrates is a dehydration of the salt, although this process resembles melting or freezing thermodynamically. At the melting point the hydrate crystals breakup into anhydrous salt and water, or into a lower hydrate and water, and the melting behaviour of the salt hydrates can be identified as congruent, incongruent and semi-congruent melting [15]. Congruent melting occurs when the anhydrous salt is completely soluble in its water of hydration at the melting temperature. Incongruent melting occurs when the

salt is not entirely soluble in its water of hydration at the melting point. Semi-congruent melting occurs when the liquid and solid phases are in equilibrium during a phase transition. The most attractive properties of salt hydrates are (i) high latent heat of fusion per unit volume, (ii) relatively high thermal conductivity (almost double of the paraffin's), and (iii) small volume changes on melting, i.e., less thermal stress effect. There is however super-cooling and phase segregation problems associated with their applications.

2.2.2. Inorganic compounds

Table 4 shows a list of available inorganic compounds but some are considered to be unsuitable for application in buildings due to their relatively small latent heat capacity. In addition many of the inorganic PCMs compounds are harmful to the environment and human health.

2.2.3. Metals

As shown in Table 5, low melting point metals and their alloys can be used as latent heat energy storage materials. These metals are have high thermal conductivity, good electrical conductivity, low vapour pressure, low heat of fusion per unit weight but high heat of fusion per unit volume (due to large density), and small volume change during phase transition [17]. Thus, if certain metal or metal alloy is used as a PCM, the heat transfer capacity will be improved significantly compared with traditional PCMs. Therefore, low melting point liquid metals could be used for application in laser system [18], USB flash memory [19] and smartphone [20] cooling.

2.3. Eutectic PCMs

Table 6 shows a list of solid–liquid eutectic PCMs. They normally consist of two or more low melting temperature components, each of which melts and freezes congruently to form a mixture of the components' crystals during crystallization [2]. The main advantage of eutectics over other types of PCMs is that their melting points can be adjusted by combining different weight percentages of components. For example tetradecane can be combined with octadecane, docosane and hexadecane to achieve

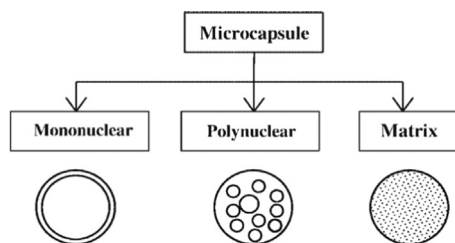


Fig. 2. Structure of MEPCMs/NEPCMs [11].

Table 1
Thermophysical properties of paraffin [2,5,9].

Name	Chemical Formula	T_m (°C)	H (kJ/kg)	k (W/m K)	ρ (kg/m ³)	C_p (kJ/kg)
<i>n</i> -Dodecane	$C_{12}H_{26}$	−12	216	0.21(s), 0.21(l)	750	n.a.
<i>n</i> -Tridecane	$C_{13}H_{28}$	−6	n.a.	n.a.	756	n.a.
<i>n</i> -Tetradecane	$C_{14}H_{30}$	4.5–5.6	231	n.a.	771	n.a.
<i>n</i> -Pentadecane	$C_{15}H_{32}$	10	207	0.17	768	n.a.
<i>n</i> -Hexadecane	$C_{16}H_{34}$	18.2	238	0.21(s)	774	n.a.
<i>n</i> -Heptadecane	$C_{17}H_{36}$	22	215	n.a.	778	n.a.
<i>n</i> -Octadecane	$C_{18}H_{38}$	28.2	245	0.35(s), 0.149(l)	814(s), 775(l)	2.14(s), 2.66(l)
<i>n</i> -Nonadecane	$C_{19}H_{40}$	31.9	222	0.21(s)	912(s), 769(l)	n.a.
<i>n</i> -Eicosane	$C_{20}H_{42}$	37	247	n.a.	n.a.	n.a.
<i>n</i> -Heneicosane	$C_{21}H_{44}$	41	215	n.a.	n.a.	n.a.
<i>n</i> -Docosane	$C_{22}H_{46}$	44	249	n.a.	n.a.	n.a.
<i>n</i> -Tricosane	$C_{23}H_{48}$	47	234	n.a.	n.a.	n.a.
<i>n</i> -Tetracosane	$C_{24}H_{50}$	51	255	n.a.	n.a.	n.a.
<i>n</i> -Pentacosane	$C_{25}H_{52}$	54	238	n.a.	n.a.	n.a.
<i>n</i> -Hexacosane	$C_{26}H_{54}$	56	257	0.21(s)	770	n.a.
<i>n</i> -Heptacosane	$C_{27}H_{56}$	59	236	n.a.	773	n.a.
<i>n</i> -Octacosane	$C_{28}H_{58}$	61	255	n.a.	910(s), 765(l)	n.a.
<i>n</i> -Nonacosane	$C_{29}H_{60}$	64	240	n.a.	n.a.	n.a.
<i>n</i> -Triacontane	$C_{30}H_{62}$	65	252	n.a.	n.a.	n.a.
<i>n</i> -Hentriacontane	$C_{31}H_{64}$	68	242	n.a.	930(s), 830(l)	n.a.
<i>n</i> -Dotriacontane	$C_{32}H_{66}$	70	170	n.a.	n.a.	n.a.
<i>n</i> -Tritriacontane	$C_{33}H_{68}$	71	189	n.a.	n.a.	n.a.
<i>n</i> -Tetraatriacontane	$C_{34}H_{70}$	75.9	269	n.a.	772.8(s)	n.a.

Legend: C_p : Specific heat (kJ/kg), H : Latent heat (kJ/kg), k : Thermal conductivity (W/m K), T_m : Melting temperature (°C), ρ : Density (kg/m³).

Table 2

Thermophysical properties of non-paraffin PCMs [2,5,6,9,14].

Name	T_m (°C)	H (kJ/kg)	k (W/m · K)	ρ (kg/m ³)	C_p (kJ/kg)
Triethylene glycol	–7	247	n.a.	1200(l)	n.a.
N-Tetradecane	5.5	226	n.a.	n.a.	n.a.
Formic acid	7.8	247	n.a.	1226.7	n.a.
Dimethyl adipate [13]	9.7	164.6	0.358	1062	n.a.
Propyl palmitate	10	186	n.a.	n.a.	n.a.
Tetrabutyl ammoniumbromide (type A–type B)	10–12	193–199	n.a.	n.a.	n.a.
Isopropyl palmitate	11	n.a.	n.a.	n.a.	n.a.
Oleic acid	13.5–16.3	n.a.	n.a.	863	n.a.
Isopropyl stearate	14–19	140–142	n.a.	n.a.	n.a.
Caprylic acid	16.3	148	0.149	901	n.a.
Dimethyl sulfoxide	16.5	85.7	n.a.	1009(l)	n.a.
Acetic acid	16.7	187–273	n.a.	1050	n.a.
Glycerin	17.9	198.7	n.a.	1260	n.a.
Butyl stearate	19	140	n.a.	n.a.	n.a.
Propyl palmitate	19	186	n.a.	n.a.	n.a.
Polyethylene glycol 600	20–25	146	n.a.	1100	n.a.
Lithium chloride ethanolate	21	188	n.a.	n.a.	n.a.
Dimethyl sabacate	21	120–135	n.a.	n.a.	n.a.
Octadecyl 3-mencaptopropylate	21	143	n.a.	n.a.	n.a.
D-Lactic acid	26	184	n.a.	1249	n.a.
Vinyl stearate	27–29	122	n.a.	n.a.	n.a.
Acid Methyl pentacosane	29	197	n.a.	n.a.	n.a.
Methyl palmitate	29	205	n.a.	n.a.	n.a.
Capric acid	32	152.7	0.153	878	n.a.
Erucic acid	33	n.a.	n.a.	853	n.a.
Trimyristin	33–57	201–213	n.a.	862	n.a.
Polyethelene glycol 900 (PEG900)	34	150.5	0.188	1100(l), 1200 (s)	2.26
Camphenilone	39	205	n.a.	n.a.	n.a.
Caprilone	40	260	n.a.	n.a.	n.a.
Docasyl bromide	40	201	n.a.	n.a.	n.a.
n-Henicosane	40.5	161	n.a.	n.a.	n.a.
Phenol	41	120	n.a.	n.a.	n.a.
Heptadecanone	41	201	n.a.	n.a.	n.a.
1-Cyclohexyloctadecane	41	218	n.a.	n.a.	n.a.
4-Heptadecanone	41	197	n.a.	n.a.	n.a.
Stearic acid	41–43	211.6	1.6	862(l) 1007(s)	2.27(l) 1.76(s)
Methyl-12-hydroxy-stearate	42–43	120–126	n.a.	n.a.	n.a.
Lauric acid	42–44	178	1.6	1007(s), 862(l)	2.27(l) 1.76(s)
n-Lauric acid	43	183	n.a.	n.a.	n.a.
p-Joluidine	43.3	167	n.a.	n.a.	n.a.
Cyanamide	44	209	n.a.	1080	n.a.
N-Docosane	44.5	157	n.a.	n.a.	n.a.
Methyl eicosanate	45	230	n.a.	851	n.a.
Elaidic acid	47	218	n.a.	851	n.a.
n-Tricosane	47.6	130	n.a.	n.a.	n.a.
Pelargonic acid	48	n.a.	n.a.	n.a.	n.a.
3-Heptadecanone	48	218	n.a.	n.a.	n.a.
2-Heptadecanone	48	218	n.a.	n.a.	n.a.
Hydrocinnamic acid	48	118	n.a.	n.a.	n.a.
Cetyl alcohol	49.3	141	n.a.	n.a.	n.a.
Camphene	50	238	n.a.	842	n.a.
a-Nephthylamine	50	93	n.a.	n.a.	n.a.
O-Nitroaniline	50	93	n.a.	n.a.	n.a.
9-Heptadecanone	51	213	n.a.	n.a.	n.a.
Thymol	51.5	115	n.a.	n.a.	n.a.
Methyl behenate	52	234	n.a.	n.a.	n.a.
Myristic acid	52.2–58	182.6–199	n.a.	862.2	n.a.
Pentadecanoic acid	52.5	178	n.a.	n.a.	n.a.
Diphenyl amine	52.9	107	n.a.	n.a.	n.a.
P-Dichlorobenzene	53.1	121	n.a.	n.a.	n.a.
N-Pentacosane	53.7	164	n.a.	n.a.	n.a.
Oxolate	54.3	178	n.a.	n.a.	n.a.
Tristearin	54.5	191	n.a.	n.a.	n.a.
Hypophosphoric acid	55	213	n.a.	n.a.	n.a.
O-Xylene dichloride	55	121	n.a.	n.a.	n.a.
Palmatic acid	55	163	n.a.	n.a.	n.a.
β Chloroacetic acid	56	147	n.a.	n.a.	n.a.
Chloroacetic acid	56	130	n.a.	1580	n.a.
N-Hexacosane	56.3	255	n.a.	n.a.	n.a.
Nitro naphthalene	56.7	103	n.a.	n.a.	n.a.
Palmitic acid	57.8–61.8	185.4	0.162	850(l), 989 (s)	n.a.
Heptaudecanoic acid	60.6	189	n.a.	n.a.	n.a.
α -Chloroacetic acid	61.2	130	n.a.	n.a.	n.a.
n-Octacosane	61.4	134	n.a.	n.a.	n.a.
Bee wax	61.8	177	n.a.	950	n.a.

Table 2 (continued)

Name	T_m (°C)	H (kJ/kg)	k (W/m · K)	ρ (kg/m ³)	C_p (kJ/kg)
Glycolic acid	63	109	n.a.	n.a.	n.a.
P-Bromophenol	63.5	86	n.a.	n.a.	n.a.
Azobenzene	67.1	121	n.a.	n.a.	n.a.
Acrylic acid	68	115	n.a.	n.a.	n.a.
Dintro toluene (2,4)	70	111	n.a.	n.a.	n.a.
Oxazoline wax-TS 970	74	n.a.	n.a.	n.a.	n.a.
Arachic acid	76.5	n.a.	n.a.	n.a.	n.a.
Phenylacetic acid	76.7	102	n.a.	n.a.	n.a.
Thiosinamine	77	140	n.a.	n.a.	n.a.
Bromcamphor	77	174	n.a.	1449	n.a.
Benzylamine	78	174	n.a.	n.a.	n.a.
Durene	79.3	156	n.a.	838	n.a.
Acetamide	81	241	n.a.	1159	n.a.
Methyl brombrenzoate	81	126	n.a.	n.a.	n.a.
Alpha naphthol	96	163	n.a.	1095	n.a.
Glutaric acid	97.5	156	n.a.	1429	n.a.
p-Xylene dichloride	100	138.7	n.a.	n.a.	n.a.
Methyl fumarate	102	242	n.a.	1045	n.a.
Catechol	104.3	207	n.a.	1370	n.a.
Quinone	115	171	n.a.	1318	n.a.
Acetanilide	115	142	n.a.	1210	n.a.
Erythritol	117.7	339.8	0.326(l), 0.733(s)	1300(l), 1480 (s)	2.61(l), 2.25(s)
Succinic anhydride	119	204	n.a.	1104	n.a.
Valporic acid	120	n.a.	n.a.	n.a.	n.a.
Benzoic acid	121.7	142.8	n.a.	1266	n.a.
Stibene	124	167	n.a.	1164	n.a.
Benzamide	127.2	169.4	n.a.	1341	n.a.
Phenacetin	137	136.7	n.a.	n.a.	n.a.
Acetyl-p-toluidene	146	180	n.a.	n.a.	n.a.
Phenylhydrazone benzaldehyde	155	134.8	n.a.	n.a.	n.a.
Salicylic acid	159	199	n.a.	1443	n.a.
Benzanilide	161	162	n.a.	n.a.	n.a.
O-Mannitol	166	294	n.a.	1489	n.a.
Hydroquinone	172.4	258	n.a.	1358	n.a.
p-Aminobenzoic acid	187	153	n.a.	n.a.	n.a.

a melting temperature range of 1.5–5.6 °C; 'Capric+lauric acid' composed at the weight percentages of 61.5%/38.5% and 45%/55% are melted at 19.1 °C and 21 °C, respectively. Eutectics do also possess high thermal conductivity and density and do not experience any segregation and supercooling. However, the latent and specific heat capacities are much smaller than salt hydrates and paraffin.

2.4. Analysis of various PCMs

As highlighted in previous sections, the advantages of organic solid–liquid PCMs include availability in a large temperature range, less super cooling, ability to melt congruently, self-nucleating, compatibility with conventional materials of construction, no segregation, chemically stable, high heat of fusion, safe and non-reactive and recyclable. They do however have disadvantages such as low thermal conductivity, low density and being flammable. On the other hand, the inorganic solid–liquid PCMs have high density that means they have higher heat storage capacity, they are cheaper and readily available, possess high thermal conductivity and are non-flammable. Nevertheless, there are also some problems such as supercooling and phase segregation associated with phase change process.

Fig. 3 illustrates the relationship between latent heat capacity of various PCMs and their melting point temperatures. It shows that the organic, salt hydrate, eutectic and solid–solid PCMs have relatively lower melting temperatures. The inorganic compound and metallic PCMs have much higher melting temperature range but the metallic PCMs do in general have the lowest latent energy storage capacity. Tables 1–5 indicate that only about 10% of thermal conductivity (k) data for PCMs have been established over the past few years, which distributed in the range of

0.149–61 W/m K. In order to highlight the difference of various kinds of PCMs the relationship between the $\log(k)$ and the melting temperatures is presented in Fig. 4, and it is clear that the metallic and their compounds have much higher thermal conductivities than the other inorganic PCMs and organic PCMs by over 10 W/m K ($\log(k) > 1$). On the other hand, most of the inorganic PCMs have higher thermal conductivity than the organic PCMs.

3. Development of micro-/nano-encapsulated PCMs

Microencapsulation is defined as a process in which tiny particles or droplets are surrounded by a coating material, or embedded in a homogeneous or heterogeneous matrix, in order to provide small capsules with useful properties [21]. Microencapsulation processes are usually categorized into two groups: physical processes and chemical processes. Physical methods include spray cooling, spray drying and fluidized bed processes. However, physical methods are limited by their granulated sizes thus making them useful for producing microencapsulated PCM particles [22]. On the other hand, chemical methods can produce much smaller encapsulated PCM particles. This section therefore reviews various types of fabrication technologies for micro-/nano-encapsulated PCMs (MEPCM/NEPCM).

3.1. In-situ polymerization

In-situ polymerization involves a process whereby chemical reaction takes place between two immiscible liquids (water soluble phase and oil soluble phase) in a continuous phase, such as emulsion/mini-emulsion, suspension, and interfacial polycondensations [11]. In general, the processes usually contains 4 steps: (1) oil/water (O/W) emulsion production, (2) preparation of prepolymer

Table 3
Thermophysical properties of salt hydrates [2, 5, 9, 16].

Name	Chemical formula	T_m (°C)	H (kJ/kg)	k (W/m K)	ρ (kg/m ³)	C_p (kJ/kg)
Lithium chlorate trihydrate	$\text{LiClO}_3 \cdot 3\text{H}_2\text{O}$	8	253	n.a.	1720(s), 1530(l)	n.a.
Zinc chloride trihydrate	$\text{ZnCl}_2 \cdot 3\text{H}_2\text{O}$	10	n.a.	n.a.	n.a.	n.a.
Ammonium chloride Sodium sulphate decahydrate	$\text{NH}_4\text{Cl} \cdot \text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	11	163	n.a.	n.a.	n.a.
Dipotassium hydrogen phosphate hexahydrate	$\text{K}_2\text{HPO}_4 \cdot 6\text{H}_2\text{O}$	14	109	n.a.	n.a.	n.a.
Sodium chloride Sodium sulphate decahydrate	$\text{NaCl} \cdot \text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	18	286	n.a.	n.a.	n.a.
Potassium fluoride tetrahydrate	$\text{KF} \cdot 4\text{H}_2\text{O}$	18	330	n.a.	n.a.	n.a.
Dipotassium hydrogen phosphate tetrahydrate	$\text{K}_2\text{HPO}_4 \cdot 4\text{H}_2\text{O}$	18.5	231	n.a.	1447(l), 1480(s)	1.84(s), 2.39(l)
Iron bromide hexahydrate	$\text{FeBr}_3 \cdot 6\text{H}_2\text{O}$	21	105	n.a.	n.a.	n.a.
Manganese nitrate hexahydrate	$\text{Mn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	25.5	125.9–148	n.a.	1738(s), 1728(l)	n.a.
Lithium metaborate octahydrate	$\text{LiBO}_2 \cdot 8\text{H}_2\text{O}$	25.7	289	n.a.	n.a.	n.a.
Calcium chloride hexahydrate	$\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$	29–30	170–192	1.008(s), 0.561(l)	1802(s), 1562(l)	n.a.
Calcium chloride dodecahydrate	$\text{CaCl}_2 \cdot 12\text{H}_2\text{O}$	29.8	174	1.09(s), 0.53(l)	1710(s), 1530(s)	1.4(s), 2.2(l)
Lithium nitrate trihydrate	$\text{LiNO}_3 \cdot 3\text{H}_2\text{O}$	30	189–296	n.a.	n.a.	n.a.
Lithium nitrate dihydrate	$\text{LiNO}_3 \cdot 2\text{H}_2\text{O}$	30	296	n.a.	n.a.	n.a.
Sodium sulphate decahydrate	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	32	251–254	0.544	1485(s)	2
Sodium carbonate decahydrate	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	32	267	0.514(s), 0.224(l)	830	1.92(s), 3.26(l)
Iron potassium alum	$\text{KFe}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$	33	173	n.a.	n.a.	n.a.
Calcium bromide hexahydrate	$\text{CaBr}_2 \cdot 6\text{H}_2\text{O}$	34	115–138	n.a.	2194(s), 1956(l)	n.a.
Lithium bromide dihydrate	$\text{LiBr} \cdot 2\text{H}_2\text{O}$	34	124	n.a.	n.a.	n.a.
Dipotassium hydrogen phosphate dodecahydrate	$\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$	35–40	256–281	n.a.	1522	n.a.
Zinc nitrate hexahydrate	$\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	36	134–147	0.464	1937(s), 1828(l)	n.a.
Manganese nitrate tetrahydrate	$\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	37	115	n.a.	n.a.	n.a.
Iron chloride hexahydrate	$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	37	223	n.a.	n.a.	n.a.
Calcium chloride tetrahydrate	$\text{CaCl}_2 \cdot 4\text{H}_2\text{O}$	39	158	n.a.	n.a.	n.a.
Copper sulphate heptahydrate	$\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$	40.7	171	n.a.	n.a.	n.a.
Potassium fluoride dihydrate	$\text{KF} \cdot 2\text{H}_2\text{O}$	42	162–266	n.a.	n.a.	n.a.
Magnesium iodide octahydrate	$\text{MgI}_2 \cdot 8\text{H}_2\text{O}$	42	133	n.a.	n.a.	n.a.
Calcium iodide hexahydrate	$\text{CaI}_2 \cdot 6\text{H}_2\text{O}$	42	162	n.a.	n.a.	n.a.
Calcium nitrate tetrahydrate	$\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	43–47	106–140	n.a.	n.a.	n.a.
Zinc nitrate tetrahydrate	$\text{Zn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	45	110	n.a.	n.a.	n.a.
Tripotassium phosphate tribasic heptahydrate	$\text{K}_3\text{PO}_4 \cdot 7\text{H}_2\text{O}$	45	145	n.a.	n.a.	n.a.
Dipotassium hydrogen phosphate heptahydrate	$\text{K}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$	45	145	n.a.	n.a.	n.a.
Iron nitrate nonahydrate	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	47	155–190	n.a.	n.a.	n.a.
Magnesium nitrate tetrahydrate	$\text{Mg}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	47	142	n.a.	n.a.	n.a.
Sodium sulphite pentahydrate	$\text{Na}_2\text{SiO}_3 \cdot 5\text{H}_2\text{O}$	48	168	n.a.	n.a.	n.a.
Sodium sulphite tetrahydrate	$\text{Na}_2\text{SiO}_3 \cdot 4\text{H}_2\text{O}$	48	168	n.a.	n.a.	n.a.
Dipotassium hydrogen phosphate heptahydrate	$\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$	48	135–170	n.a.	n.a.	n.a.
Sodium thiosulfate pentahydrate	$\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$	48	209	n.a.	1600	n.a.
Dipotassium hydrogen phosphate trihydrate	$\text{K}_2\text{HPO}_4 \cdot 3\text{H}_2\text{O}$	48	99	n.a.	n.a.	n.a.
Magnesium sulphate heptahydrate	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	48.5	202	n.a.	n.a.	n.a.
Calcium nitrate trihydrate	$\text{Ca}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	51	104	n.a.	n.a.	n.a.
Sodium nitrate hexahydrate	$\text{Na}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	53	158	n.a.	n.a.	n.a.
Zinc nitrate dihydrate	$\text{Zn}(\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$	55	68	n.a.	n.a.	n.a.
Iron chloride dihydrate	$\text{FeCl}_3 \cdot 2\text{H}_2\text{O}$	56	90	n.a.	n.a.	n.a.
Cobaltous nitrate hexahydrate	$\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	57	115	n.a.	n.a.	n.a.
Nick nitrate hexahydrate	$\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	57	169	n.a.	n.a.	n.a.
Manganese chloride tetrahydrate	$\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$	58	151	n.a.	n.a.	n.a.
Sodium acetate trihydrate	$\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$	58	270–290	n.a.	1450	n.a.
Lithium acetate dihydrate	$\text{LiC}_2\text{H}_3\text{O}_2 \cdot 2\text{H}_2\text{O}$	58	251–377	n.a.	n.a.	n.a.
Magnesium chloride tetrahydrate	$\text{MgCl}_2 \cdot 4\text{H}_2\text{O}$	58	178	n.a.	n.a.	n.a.
Sodium hydroxide monohydrate	$\text{NaOH} \cdot \text{H}_2\text{O}$	58	272	n.a.	n.a.	n.a.
Cadmium nitrate tetrahydrate	$\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$	59	98	n.a.	n.a.	n.a.
Cadmium nitrate monohydrate	$\text{Cd}(\text{NO}_3)_2 \cdot \text{H}_2\text{O}$	59.5	107	n.a.	n.a.	n.a.
Iron nitrate hexahydrate	$\text{Fe}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	60.5	126	n.a.	n.a.	n.a.
Sodium aluminium sulphate dodecahydrate	$\text{NaAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$	61	181	n.a.	n.a.	n.a.
Sodium aluminium sulphate decahydrate	$\text{NaAl}(\text{SO}_4)_2 \cdot 10\text{H}_2\text{O}$	61	181	n.a.	n.a.	n.a.
Ferrous sulphate heptahydrate	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	64	200	n.a.	n.a.	n.a.
Sodium phosphate dodecahydrate	$\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$	65	190	n.a.	n.a.	n.a.
Sodium borate decahydrate	$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$	68	n.a.	n.a.	n.a.	n.a.
Lithium ethanoate dihydrate	$\text{LiCH}_3\text{COO} \cdot 2\text{H}_2\text{O}$	70	150–251	n.a.	n.a.	n.a.
Sodium polyphosphate decahydrate	$\text{Na}_2\text{P}_2\text{O}_7 \cdot 10\text{H}_2\text{O}$	70	186–230	n.a.	n.a.	n.a.
Aluminium nitrate nonahydrate	$\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	72	155–176	n.a.	n.a.	n.a.
Barium hydroxide octahydrate	$\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$	78	265–280	1.255(s), 0.653(l)	2070(s), 1937(l)	n.a.
Aluminium sulphate 18-hydrate	$\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$	88	218	n.a.	n.a.	n.a.
Strontium hydroxide octahydrate	$\text{Sr}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$	89	370	n.a.	n.a.	n.a.
Magnesium nitrate hexahydrate	$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	89.9	167	0.490(l), 0.611(s)	1550(l), 1636(s)	n.a.
Aluminium potassium sulphate dodecahydrate	$\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$	91	184	n.a.	n.a.	n.a.
Ammonium alum	$(\text{NH}_4)\text{Al}(\text{SO}_4) \cdot 6\text{H}_2\text{O}$	95	269	n.a.	n.a.	n.a.
Lithium chloride monohydrate	$\text{LiCl} \cdot \text{H}_2\text{O}$	99	212	n.a.	n.a.	n.a.
Calcium bromide tetrahydrate	$\text{CaBr}_2 \cdot 4\text{H}_2\text{O}$	110	n.a.	n.a.	n.a.	n.a.
Aluminium sulphate 16-hydrate	$\text{Al}_2(\text{SO}_4)_3 \cdot 16\text{H}_2\text{O}$	112	n.a.	n.a.	n.a.	n.a.
Magnesium chloride hexahydrate	$\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$	117	167	0.570(l), 0.704(s)	1450(l), 1570(s)	2.61(l), 2.25(s)

mixture liquid, (3) adding prepolymer mixture liquid into O/W emulsion to encapsulate core material particles, and (4) washing and drying MEPCM/NEPCM. As an example, Fig. 5 shows the process of encapsulation of *n*-octadecane with resorcinol-modified melamine–formaldehyde shell [23] using the in-situ polymerization process.

Choi et al. [24] used a melamine formaldehyde shell and 5 wt% styrene–maleic anhydride–monomethyl (SMA) as emulsifier to encapsulate tetradecane. The result showed reduction in the size of the capsules and uniformity was improved with an optimum emulsion speed of up to 8000 rpm. Fang et al. [25] used a UF shell to encapsulate tetradecane PCM. The liquid was emulsified at 60 °C for 30 min with an emulsion speed of 1500 rpm. Sodium dodecyl sulfate and resorcin were used as emulsifier and system modifier, respectively. Further analysis of the results showed that the thermal stability of the capsules was also improved after adding

2–5 wt% of NaCl. Yang et al. [26] successfully fabricated four types of MEPCM with different shell materials (PVAc, PS, PMMA and PEMA). Jin et al. [27] dispersed paraffin using hydrolyzed-styrene-alt-maleic anhydride (HSMA) as the emulsifier at a speed of 12,000 rpm and then used UF as a shell material to produce MEPCM. The content of the shell material was then increased to 28% and was subjected to number of thermal cycles. The results showed the capsules to be intact thus making them thermally stable than a bulk paraffin material. Zhang et al. [28] encapsulated *n*-octadecane with MF material, emulsifier (TA 0.6–2.3 wt%) and cyclohexane (0–36.8 wt%) material. They observed that the diameters of the microcapsules could be reduced by increasing the stirring speed from 3000 rpm to 9000 rpm. Li et al. [29] emulsified *n*-octadecane at a stirring rate of 8000 rpm and then encapsulated it with MF shell to obtain MEPCM of an average diameter of 2.2 μm and latent heat capacity of 144 kJ/kg. Hong and Park [30] fabricated MEPCM by using 53 wt% fragrant margin oil and MF as core and shell materials, respectively. The experimental study, which was conducted at an emulsion speed of 3000 rpm, resulted in a MEPCM of particle size smaller than 10 μm. Other encapsulation process involving binary mixture of *n*-hexadecane and *n*-eicosane and MF as a shell material achieved a maximum energy storage capacity of 163–170 kJ/kg [31]. Song et al. [32] also introduced nano-silver particles in in-situ polymerization process to encapsulate bromo-hexadecane (BrC₁₆) with aminoplast as the shell material. Their result did prove that nano-silver particles can enhance mechanical strength and thermal stabilities of MEPCMs without experiencing any particle agglomeration problem.

3.1.1. Interfacial polycondensation

Interfacial polycondensation is one of the in-situ processes wherein a microcapsule wall of a polymer is formed at an interface between two phases with each of them containing a suitable reaction monomer. Initially, a multifunctional monomer is dissolved in the organic core material and then the resulting mixture is dispersed in an aqueous phase containing a mixture of emulsifiers and protective colloid stabilizers. Combination of monomers is then added to the aqueous phase for the formation of a polymer shell [33]. Fig. 6 shows a process used by Zhang et al. [34] to

Table 4
Thermal physical properties of inorganic compounds [2, 5].

Name	Chemical formula	T_m (°C)	H (kJ/kg)	k (W/m K)
Water	H ₂ O	0	333	0.598
Phosphorus oxychloride	POCl ₃	1	85	n.a.
Deuterium oxide	D ₂ O	3.7	318	0.595
Antimony chloride	SbCl ₅	4	33	n.a.
Sulphuric acid	H ₂ SO ₄	10.4	100	0.26
Iodine chloride (β)	ICl (β)	13.9	56	n.a.
Molybdenum fluoride	MoF ₆	17	50	n.a.
Sulphur trioxide (α)	SO ₃ (α)	17	108	n.a.
Iodine chloride (α)	ICl (α)	17.2	69	n.a.
Tetraphosphorus hexaoxide	P ₄ O ₆	23.7	64	n.a.
Phosphoric acid	H ₃ PO ₄	26	147	0.434[16]
Arsenic bromide	AsBr ₃	30	38	n.a.
Tin bromide	SnBr ₄	30	28	0.0859
Boron iodide	BI ₃	31.8	10	n.a.
Sulphur trioxide (β)	SO ₃ (β)	32.3	151	n.a.
Titanium bromide	TiBr ₄	38.2	23	n.a.
Hypophosphoric acid	H ₄ P ₂ O ₆	55	213	n.a.
Sulphur trioxide (γ)	SO ₃ (γ)	62.1	331	n.a.
Antimony trichloride	SbCl ₃	73.4	25	n.a.
Sodium nitrate	NaNO ₃	307	199	0.51
Potassium nitrate	KNO ₃	380	266	0.45
Potassium hydroxide	KOH	380	149	n.a.

Table 5
Melting temperature and latent heat of metallic PCMs [2,17].

Name	T_m (°C)	H (kJ/kg)	k (W/m K)	ρ (kg/m ³)	C_p (kJ/kg)
Mercury	−38.87	11.4	8.34	13,546	0.139
Cesium	28.65	16.4	17.4	1796	0.236
Gallium–gallium antimony	29.8	n.a.	n.a.	n.a.	n.a.
Gallium	30	80.3	29.4	5907	0.37
Rubidium	38.85	25.74	29.3	1470	0.363
Bismuth–lead–indium–tin–cadmium (Bi _{44.7} Pb _{22.6} In _{19.1} Sn _{8.3} Cd _{5.3})	47	36.8	15	9160	0.197
Bismuth–indium–lead–tin (Bi ₄₉ In ₂₁ Pb ₁₈ Sn ₁₂)	58	28.9	10	9010	0.201
Cerrolow	58	90.9	n.a.	n.a.	n.a.
Bismuth–cadmium–indium	61	25	n.a.	n.a.	n.a.
Potassium	63.2	59.59	54.0	664	0.78
Cerrobend	70	32.6	n.a.	n.a.	n.a.
Bismuth–lead–tin–cadmium (Bi ₅₀ Pb _{26.7} Sn _{13.3} Cd ₁₀)	70	39.8	18	9580	0.184
Bismuth–lead–indium	70	29	n.a.	n.a.	n.a.
Bismuth–indium	72	25	n.a.	n.a.	n.a.
Bismuth–lead–tin (Bi ₅₂ Pb ₃₀ Sn ₁₈)	96	34.7	24	9600	0.167
Bismuth–lead–tin	96	n.a.	n.a.	n.a.	n.a.
Sodium	97.8	n.a.	n.a.	n.a.	n.a.
Bi–Pb	125	n.a.	n.a.	n.a.	n.a.
Bismuth–tin (Bi ₅₈ Sn ₄₂)	138	44.8	19	8560	0.201
Indium	156.8	28.59	36.4	7030	0.23
Lithium	186	433.78	41.3	515	4.389
Tin–zinc (Sn ₉₁ Zn ₉)	199	32.5	61	7270	0.272
Tin	232	60.5	15.08	730	0.221
Bismuth	271.4	53.3	8.1	979	0.122

Table 6
Thermophysical properties of eutectic PCMs [2,6,9,17].

Name	Composition (wt%)	T_m (°C)	H (kJ/kg)
Diethylene glycol	n.a.	−10	247
Tetradecane + octadecane	n.a.	−4.02	227.52
Water + polyacrylamide	n.a.	0	295
Tetradecane + docosane	n.a.	1.5–5.6	234.33
Tetradecane + hexadecane	91.67 + 8.33	1.7	156.2
Tetradecane + geneicosane	n.a.	3.54–	200.28
		5.56	
$\text{Na}_2\text{SO}_4 + \text{NaCl} + \text{KCl} + \text{H}_2\text{O}$	31 + 13 + 16 + 40	4	234
Tetrahydrofuran (THF)	n.a.	5	280
Pentadecane + heneicosane	n.a.	6.23–7.21	128.25
Pentadecane + docosane	n.a.	7.6–8.99	214.83
Pentadecane + octadecane	n.a.	8.5–9.0	271.93
$\text{Na}_2\text{SO}_4 + \text{NaCl} + \text{NH}_4\text{Cl} + \text{H}_2\text{O}$	32 + 14 + 12 + 42	11	n.a.
$\text{C}_5\text{H}_5\text{C}_6\text{H}_5 + (\text{C}_6\text{H}_5)_2\text{O}$	26.5 + 73.5	12	97.9
Triethylolethane + water + urea	38.5 + 31.5 + 30	13.4	160
$\text{CaCl}_2 \cdot 6\text{H}_2\text{O} + \text{CaBr}_2 \cdot 6\text{H}_2\text{O}$	45 + 55	14.7	140
$\text{Na}_2\text{SO}_4 + \text{NaCl} + \text{H}_2\text{O}$	37 + 17 + 46	18	n.a.
Capric + lauric acid	61.5 + 38.5	19.1	132
Capric + lauric acid	82 + 18	19.1–20.4	147
Capric + lauric acid	45 + 55	21	143
Capric + myristic	73.5 + 26.5	21.4	152
Capric + palmitate	75.2 + 24.8	22.1	153
$\text{Na}_2\text{S}_4 + \text{MgSO}_4 + \text{H}_2\text{O}$	25 + 21 + 54	24	n.a.
$\text{C}_{14}\text{H}_{28}\text{O}_2 + \text{C}_{10}\text{H}_{20}\text{O}_2$	34 + 66	24	147.7
$\text{CaCl}_2 \cdot 6\text{H}_2\text{O} + \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$	50 + 50	25	95
$\text{CaCl}_2 \cdot 6\text{H}_2\text{O} + \text{Nucleat} + \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$	66.7 + 33.3	25	127
$\text{CaCl}_2 + \text{NaCl} + \text{KCl} + \text{H}_2\text{O}$	48 + 4.3 + 0.4 + 47.3	26.8	188
Capric + stearate	86.6 + 13.4	26.8	160
$\text{CH}_3\text{CONH}_2 + \text{NH}_2\text{CONH}_2$	50 + 50	27	163
Triethylolethane + urea	62.5 + 37.5	29.8	218
$\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O} + \text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	47 + 53	30	136
$\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O} + \text{NH}_2\text{CONH}_2$	40 + 60	30	200.5
Lauric + palmitic acid	69 + 31	35.2	166.3
$\text{NH}_2\text{CONH}_2 + \text{NH}_4\text{NO}_3$	53 + 47	46	95
$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{NH}_4\text{NO}_3$	61.5 + 38.5	52	125.5
$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$	58.7 + 41.3	59	132.2
$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$	50 + 50	59.1	144
$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	53 + 47	61	148
$\text{CH}_3\text{CONH}_2 + \text{C}_{17}\text{H}_{35}\text{COOH}$	50 + 50	65	218
$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{MgBr}_2 \cdot 6\text{H}_2\text{O}$	59 + 41	66	168
Napthalene + benzoic acid	67.1 + 32.9	67	123.4
$\text{AlCl}_3 + \text{NaCl} + \text{ZrCl}_2$	79 + 17 + 4	68	234
$\text{AlCl}_3 + \text{NaCl} + \text{KCl}$	66 + 20 + 14	70	209
$\text{NH}_2\text{CONH}_2 + \text{NH}_4\text{Br}$	66.6 + 33.4	76	151
$\text{LiNO}_3 + \text{NH}_4\text{NO}_3 + \text{NaNO}_3$	25 + 65 + 10	80.5	113
$\text{LiNO}_3 + \text{NH}_4\text{NO}_3 + \text{KNO}_3$	26.4 + 58.7 + 14.9	81.5	116
$\text{LiNO}_3 + \text{NH}_4\text{NO}_3 + \text{NH}_4\text{Cl}$	27 + 68 + 5	81.6	108
$\text{AlCl}_3 + \text{NaCl} + \text{KCl}$	60 + 26 + 14	93	213
$\text{AlCl}_3 + \text{NaCl}$	66 + 34	93	201
$\text{NaNO}_2 + \text{NaNO}_3 + \text{KNO}_3$	40 + 7 + 53	142	n.a.

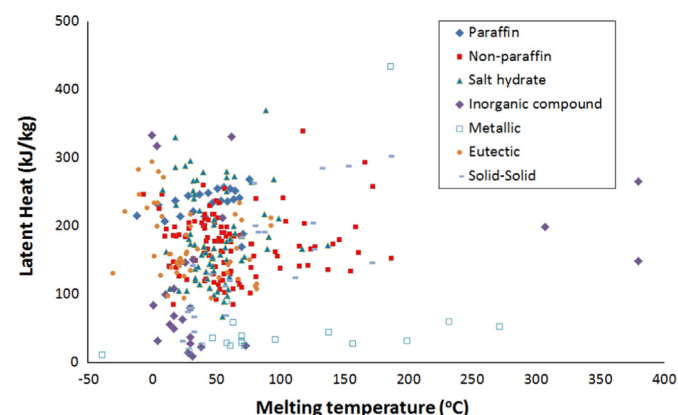


Fig. 3. Melting temperature and latent heat distribution for different types of PCMs.

synthesize microencapsulated *n*-octadecane with polyurea shells, which were produced with tolylene 2,4-diisocyanate (TDI) and three different amines namely, ethylene diamine (EDA), diethylene triamine (DETA) and Jeffamine T403. The oil solution was prepared by mixing together TDI and *n*-octadecane and then adding water to the mixture at a stirring speed of 3000 rpm to produce O/W emulsion. Furthermore, an EDA solution consisting of 0.1 wt% SMA was added into O/W emulsion at a stirring rate of 600 rpm to complete the interfacial polymerization process. The MEPCM produced with Jeffamine as the amine monomer displayed the best anti-osmosis property.

Other studies have also been conducted using the same process. For instance, Chen et al. [35] used butyl stearate as core material and TDI/EDA as shell material to produce MEPCM. They also investigated the effect of different stirring rates (300–700 rpm), emulsifier (polyethylene glycol octylphenyl ether (OP)-10) content and core/shell mass ratios on the quality of the MEPCM. The results established a suitable stirring rate of 500 rpm for a core/shell mass ratio of 4:1 but found no significant influence on the emulsifier. Zou et al. [36] also encapsulated hexadecane with the same type of shell and emulsifier material as above but at a stirring rate of 300 rpm. Cho [37], Siddhan [38] and Su [39] all used TDI and DETA to produce shell materials for encapsulating *n*-octadecane. The main difference between their methods was that Siddhan and Su used higher stirring speed ranging from 2500–4000 rpm to produce the O/W emulsion, whereas Cho's emulsion speed was limited to a lower speed of 300 rpm. Surprisingly, the average diameters of MEPCM particles produced by Siddhan and Su were 5 μm and 7.3 μm , respectively, as compared with 1 μm obtained by Cho. Tseng et al. [40] fabricated MEPCM of mean capsule diameter of 47–150 μm by using 53–61 wt% paraffins (*n*-pentadecane, *n*-eicosane and paraffin wax) as core materials and UF for the shell.

In order to enhance thermal conductivities of organic MEPCMs other researchers have tried the use of inorganic shell materials under the interfacial polycondensation encapsulation process. Zhang et al. [41] used silicon dioxide (SiO_2) to encapsulate *n*-octadecane and achieved thermal of about 0.6547 W/m K. Pan et al. [42] used aluminium hydroxide ($\text{Al}(\text{OH})_3$) to encapsulate palmitic acid (PA) and obtained thermal conductivities ranging between 0.7 and 0.84 W/m K. Li et al. [43] reported another development also based on silicon dioxide as the shell and paraffin as the core material. The evaluation results showed that the capsules were able to maintain stable phase transition without any sign of leakage after a number of repeated melting–freezing cycles.

3.1.2. Suspension polymerization

The use of suspension polymerization process for the fabrication of MEPCM usually follows certain basic procedures: (1) dissolution of polymer monomer into organic phase (core materials); (2) production of oil/water (O/W) emulsion; (3) separation and precipitation of the monomer molecules from core materials and generation of solid shell. Various researchers have produced MEPCMs using the suspension polymerization process. Fig. 7 describes the suspension polymerization process which Ai et al. [44] used for microencapsulating *n*-hexadecane. The results demonstrated that higher emulsion speed could reduce particle sizes of MEPCMs.

Sánchez et al. [45] fabricated MEPCMs by encapsulating 50 wt% of paraffin wax PRS, tetradecane, Rubitherm 20 and nonadecane as core materials in a polystyrene shell. Further experimental studies by Sánchez et al. [46] proved that crosslinking reaction temperature has no significant effect on the particle size distribution. It also revealed that it is difficult to achieve encapsulation when

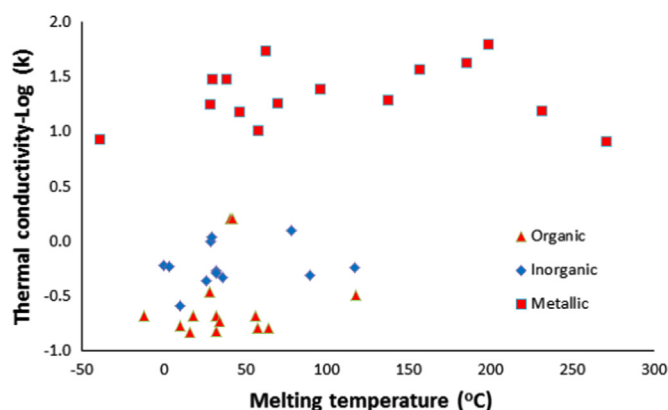


Fig. 4. Thermal conductivity distribution for different types of PCMs.

core/shell mass ratio is greater than 2.00. You et al. [47,48] produced *n*-Octadecane microcapsules with a styrene (St)-divinylbenzene (DVB) copolymer shell and achieved an average diameter of 80 μm and latent heat fusion capacity of 126 kJ/kg. Thermogravimetric (TG) analysis of the St-DVB shell showed the initial weight-loss temperatures to be above 230 $^{\circ}\text{C}$ thus making it better than MF shell. Li et al. [49] microencapsulated *n*-octadecane with styrene-1,4-butylene glycol diacrylate copolymer (PSB), styrene-divinylbenzene copolymer (PSD), styrene-divinylbenzene-1,4-butylene glycol diacrylate copolymer (PSDB), and polydivinylbenzene (PDVB) shell materials. They then analysed the morphology of the MEPCMs from scanning electron microscope (SEM) images and found the type with PSDB shell to be the best of all the samples.

Much smaller MEPCM/NEPCM capsules have also been produced by the microsuspension polymerization process. As shown in Fig. 8 [50] *n*-octadecane was encapsulated with a PDVB shell to obtain an average size of 1.5 μm MEPCM. Qiu et al. [51] produced nanoencapsulated *n*-octadecane with different polymer shells (1,4-butylene glycol diacrylate (BDDA), divinyl benzene (DVB), trimethylol propane triacrylate (TMPTA) and pentaerythritol tetraacrylate (PETRA)). Their thermal properties, thermal resistant temperatures and shell mechanical strength of capsules were enhanced by increasing the level of cross-linking agents. Simultaneously, the cross-linking agent PETRA created the best NEPCM product, which contained 75.3 wt% core material with phase change enthalpy of 156.4–182.8 kJ/kg and average particle diameter of 720 nm. Cheng et al. [52] fabricated NEPCMs with a polyurethane shell and achieved a latent heat value of 104 kJ/kg. The particle size was changed from 300 to 600 nm, whilst the crosslinking agent tripropylene glycol diacrylate (TPGDA) increased from 10 wt% to 20 wt%.

Unlike other encapsulation technologies a special precaution must be taken under some suspension polymerization processes. For instance, when using polystyrene, styrene-methyl methacrylate (St-MMA), polydivinylbenzene (PDVB) and polyurethane as MEPCM shell materials nitrogen gas protection facility would be necessary during the fabrication process and for the fact that the reaction time could take as long as 5–24 h.

3.1.3. Emulsion/mini-emulsion polymerization

Emulsion polymerization takes place over a number of steps, where various chemical and physical events take place simultaneously during the process of particle formation and growth. Three major mechanisms for particle formation have been proposed to date. Particle formation is considered to have taken place when (1) a free radical in the aqueous phase enters a monomer-swollen emulsifier micelle and propagation proceeds therein or; (2) the chain length of a free radical growing in the aqueous phase exceeds its solubility limit and precipitates to form a particle

nucleus, or; (3) a free radical growing in the aqueous phase enters a monomer droplet and propagation proceeds therein [53].

Polymethylmethacrylate (PMMA) is a common shell material used for producing MEPCM with emulsion polymerization method. For instance, Alkan et al. [54] encapsulated docosane with PMMA for thermal energy storage material. The capsules contained 28 wt% of core material with a narrow particle size distribution (average particle size was 0.16 μm) and were carefully controlled at an emulsion stirring rate of 2000 rpm. Alkan et al. [55] further coated PMMA shell material containing capsules of average diameter 0.70 μm with 35 wt% of *n*-Eicosane. Sari et al. [56,57] also produced smaller sizes of NEPCM ranging from 0.14 to 0.40 μm with 43 wt% of *n*-Octadecane and 38 wt% *n*-Heptadecane as core materials and at a stirring speed of 2000 rpm. Ma et al. [58] produced higher core material content MEPCM which the core material content to be 61.2 wt% and its latent heat capacity as much as 101 kJ/kg. During the polymerization process, UV irradiation light intensity and exposure time were used to control polymerization speed which helped to minimise the polymerization time to only 30 min. SEM analysis revealed the diameter range of the MEPCM as 0.5–2 μm . Alay et al. [59] developed *n*-hexadecane microcapsules for textile application. The mean particle diameter of the capsules was in the range of 0.22–1.05 μm with corresponding enthalpy of 68.9–145.6 kJ/kg.

Mini-emulsion polymerization is the same process as emulsion polymerization, except that smaller droplets can be produced as shown in Fig. 9. Homogenization can be achieved using an ultrasonifer (for laboratory-scale batch process) or a high-pressure homogenizer (for larger-scale processes). Luo and Zhou [60] used this method to develop nanoencapsulated paraffin as thermal energy storage material. They however found out that the thermodynamic factors (the level and type of surfactant and hydrophilic comonomer, and the monomer/paraffin ratio), kinetic factors (the level of the crosslinking agent or chain-transfer agent), and nucleation modes did have significant influence on the encapsulation process. Chen et al. [61] produced nanocapsules of an average particle size of 150 nm with *n*-Dodecanol as the core material and PMMA as the shell material. Maximum phase change enthalpy of 98.8 J/g and encapsulation efficiency of 82.2% were obtained with 3% mass ratios of polymerizable emulsifier (DNS-86)/core material and 2% co-emulsifier (hexadecane (HD))/core material. Li et al. [62] successfully synthesized NEPCM with the two-step miniemulsion polymerization method and achieved a mean particle diameter of 270 nm. The experimental results did reveal that by increasing the amount of surfactant material (sodium dodecyl sulphate (SDS)) the phase-change enthalpy of the nanocapsules did also increase but the mean particle size was reduced. Fuensanta et al. [63] encapsulated paraffin wax (RT 80) of particle size ranging from 52 to 112 nm with encapsulation efficiency of about 80%. Meanwhile, in comparison with raw RT80 its melting temperature was decreased by 1–7 $^{\circ}\text{C}$ but did show good thermal stability after 200 thermal cycles.

3.1.4. Shell materials

As summarised in Table 7, it is clear that various wax materials have been encapsulated with different shell materials but melamine formaldehyde (MF), urea formaldehyde (UF), polymethyl methacrylate (PMMA), polyurea and polystyrene appear to be the most widely used shell materials due to their good chemical stability and mechanical strength. They, however, have low thermal conductivity and also some of the chemicals such as formaldehyde can cause environmental and health problems during fabrication process of the MEPCM/NEPCM. Although high thermal conductivity inorganic shells, i.e., $\text{Al}(\text{OH})_3$ and SiO_2 have been used for thermal enhancement their presence did reduce the overall energy storage capacities of the phase change materials.

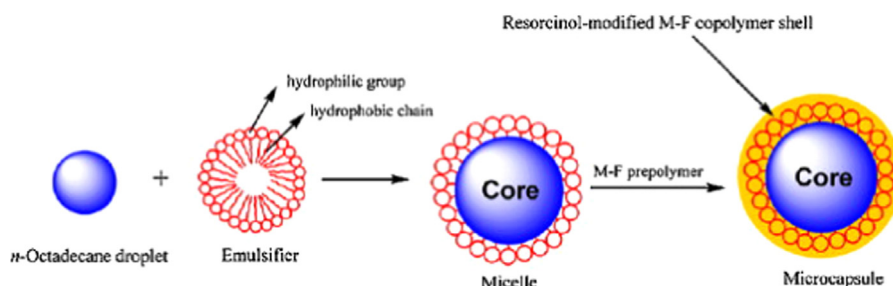


Fig. 5. Fabrication of the MEPCM by in-situ polymerization [23].

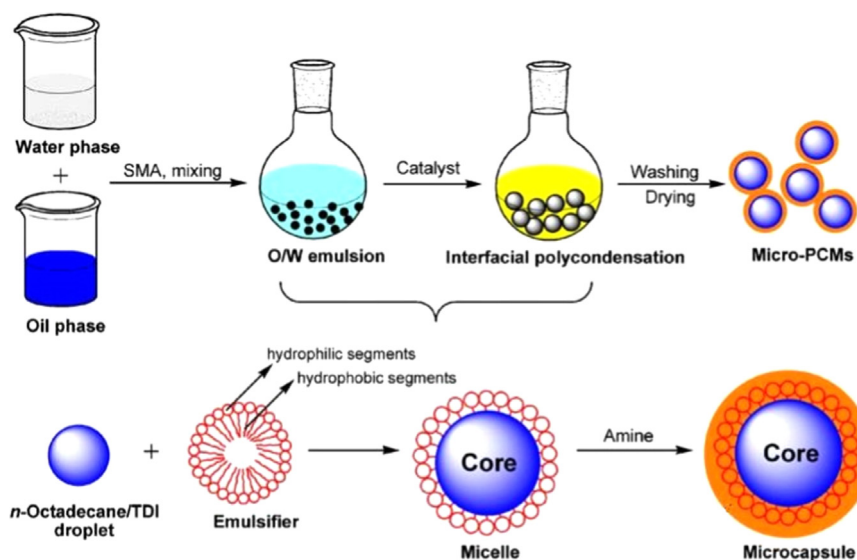


Fig. 6. Microcapsule manufactured by interfacial polycondensation [34].

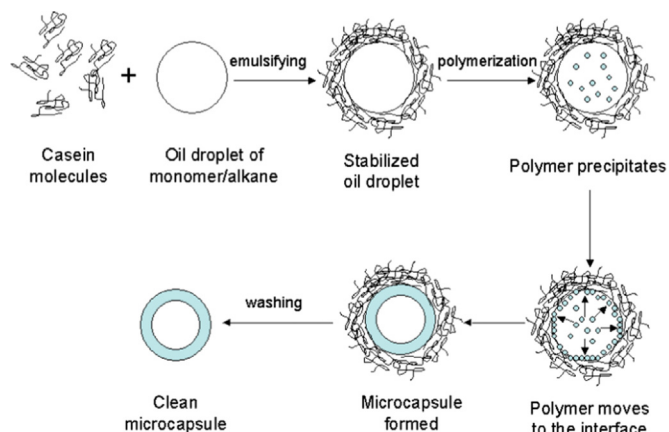


Fig. 7. Schematic of the fabrication MEPCM by suspension-like polymerization [44].

3.2. Complex coacervation

Complex coacervation involves a reaction between two or more types of polymer materials with oppositely charged crosslinks to copolymer shells. As shown in Fig. 10, the general process consists of three stages carried out under continuous agitation as follows. (1) In the first stage, coating material is dispersed in water to form a phase separation coacervation and then a core material is added into the solution to produce O/W emulsion. (2) The second stage involves adding another colloid solution charged with oppositely electric into the O/W emulsion, and then adjusting the pH number

of the solution appropriately. (3) The third stage finishes it off with the cooling down of the mixture, microencapsulation and harvesting of the MEPCM. The main limitation of this approach is the difficulty in scaling up the process [65].

Nevertheless, a number of encapsulation developments have been carried out over the past years with the complex coacervation method. Ozonur et al. [66] encapsulated natural coco-fatty acid with gelatin–gum Arabic and achieved MEPCM of particle size of 1 mm. It was subjected to a number of thermal cycles and found no physical deformation to its geometrical shape. In comparison with pure coco-fatty, its melting temperature also increased about 7 °C. Onder et al. [67] used three types of paraffin waxes: *n*-hexadecane, *n*-octadecane and *n*-nonadecane as core materials and gum arabic–gelatin mixture as a shell material to produce MEPCMs. The results showed that microcapsules produced with *n*-octadecane achieved the highest enthalpy value. They concluded that the quality of MEPCM could be improved by ensuring precise pH value and correct amount of surfactants at higher stirring rates. Hawlader et al. [68] established the optimal homogenizing time in coacervation method as 10 min and the amount of cross-linking agent required as 6–8 ml. They also demonstrated that microencapsulation efficiency is dependent upon the process parameters such as core material ratio, emulsifying time and the amount of cross-linking agent. Bayes-Garcia [69] successfully produced Rubitherm® RT 27 microcapsules from two different coacervates; sterilized gelatine/Arabic gum for the SG/AG method and Agar-Agar/Arabic Gum for the AA/AG method. According to the particle sizing analysis the average diameter of the capsules produced with the SG/AG method was 12 µm, whereas the AA/AG method achieved far smaller size of 104 nm. Su et al. [70] also used the

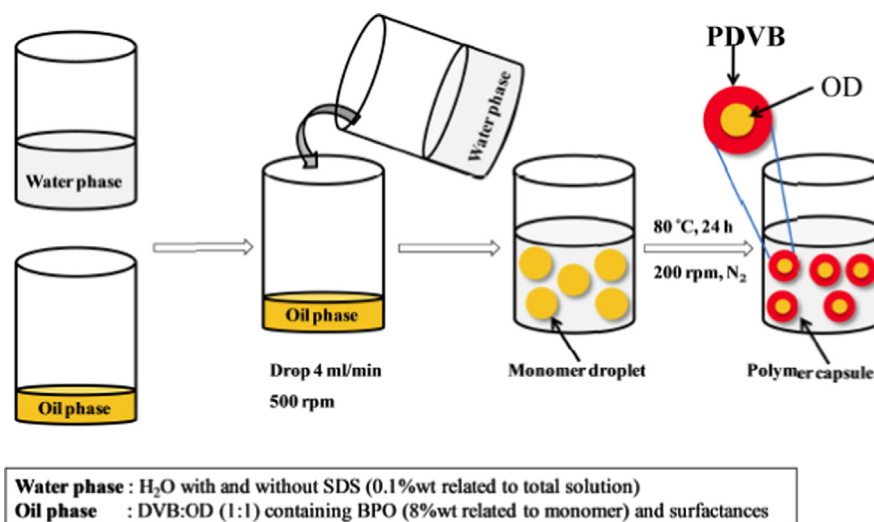


Fig. 8. Schematic of the preparation of the PDVB/Octadecane capsules by the microsuspension polymerization [50].

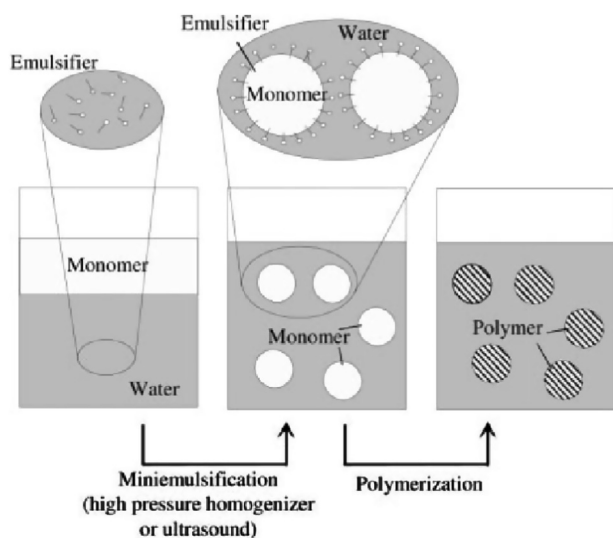


Fig. 9. The procedure of miniemulsion polymerization [64].

two-step coacervation (TSC) method and an MF shell to encapsulate *n*-octadecane. This method reduced the cracks on the shell and increased its compactness and permeability coefficient.

3.3. Sol-gel method

The term “sol-gel” is an abbreviation for “solution-gelling” and denotes a process by which largely inorganic materials are synthesized. The process follows a principle whereby a solution undergoes a transition to a gel characterized by an infinite three-dimensional network structure spreading uniformly throughout the liquid medium [11]. For instance, Zhang et al. [71] fabricated microcapsules of *n*-octadecane with silica shell to enhance thermal conductivity and phase-change performance, as shown in Fig. 11. The results achieved good phase-change performance, high encapsulation efficiency, high thermal conductivity and good antiosmosis property.

Other researchers have also fabricated MEPCMs using the sol-gel method but with silicon as a shell material. Wang et al. [72] were the first to encapsulate PCMs with silicon and to investigate the formation mechanism which established that the use of cationic surfactants (i.e., cetyltrimethylammonium chloride, dodecyltrimethylammonium chloride and dodecyltrimethylammonium bromide) as

emulsifiers are suitable for the production of MEPCMs. Li et al. [73] prepared different composite MEPCMs based on paraffin/SiO₂/expanded graphite and paraffin/SiO₂ and achieved much higher thermal conductivities than pure paraffin. Fang et al. [74] also produced paraffin-based MEPCM using SiO₂ as a shell material and reported solidification/melting temperatures of 57.02/58.37 °C and latent heat capacity of 107.05/165.68 kJ/kg. Chang et al. [75] micro-encapsulated *n*-Octadecane using a PMMA network-silica hybrid shell. They concluded that the most suitable condition for producing high percentage (74%) of PCM microcapsules of high latent heat capacity (180 kJ/kg) was to add 5% of silicon dioxide (SiO₂) to the process. Chen et al. [76] produced MEPCM (diameter 20–30 μm) capsules by using 90.7% of stearic acid (SA) and SiO₂ as core and shell materials, respectively. The experimental results achieved good thermal stability for energy storage capacity of 162.0–171.0 kJ/kg over a phase change temperature range of 52.6–53.5 °C. Chen et al. [77] further encapsulated high percentage (82.2%) content of paraffin with SiO₂ shell and obtained good thermal characteristics. The results showed that the microcapsules melted at 57.96 °C (latent heat 156.86 kJ/kg) and solidified at 55.78 °C (latent heat 144.09 kJ/kg). Sara et al. [78] synthesized PCM nanocapsules, which contained palmitic acid (PA) as core and SiO₂ as shell materials. They observed that by progressively increasing the pH level of the core material, the mean diameter and the energy storage capacity of the nanocapsules also increased. For instance an increase in pH value from 11 to 12 resulted in an increase of 183.7–722.5 nm in the particle diameter and 168–181 kJ/kg in its energy storage capacity.

3.4. Solvent extraction/evaporation method

Inorganic PCMs can be encapsulated by solvent extraction/evaporation method through the (i) incorporation of the bioactive compound, (ii) formation of the microdroplets, (iii) solvent removal and (iv) harvesting and drying the particles. As shown in Fig. 12, Salaün et al. [79] studied sodium phosphate dodecahydrate (DSP, Na₂H-PO₄·12H₂O) encapsulation by solvent evaporation-precipitation method with various organic solvents and cellulose acetate butyrate (CAB) crosslinked by methylene diisocyanate (MDI). It was evident in the results that surface appearance of MEPCM depends on the rate of polymer precipitation at the interface, and that thermal stability and the thermal degradation were strongly influenced by the core content and the surface roughness of the MEPCM.

In other studies, Wang and Huang [80] encapsulated DSP with PMMA using the same method and was able to produce uniform sizes of 1–10 μm MEPCM capsules within polymerization time of 4 h at a

Table 7
MEPCM/NEPCM fabricated by in-situ polymerization.

Core	Shell	References
Tetradecane	Melamine formaldehyde (MF)	Choi [24]
Tetradecane	Urea formaldehyde (UF)	Fang [25]
Tetradecane	(1) PVAc (polyvinyl acetate), (2) PS (poliesticrene), (3) Polymethyl methacrylate (PMMA), (4) Polyethyl methacrylate (PEMA)	Yang [26]
Paraffin	UF	Jin [27]
<i>n</i> -Octadecane	UMF/UF/MF	Zhang [65]
<i>n</i> -Octadecane	MF	Li [29], Zhang [23], Zhang [28]
Fragrant margin oil	MF	Hong [30]
<i>n</i> -Hexadecane and <i>n</i> -eicosane	MF	Salaün [31]
<i>n</i> -Octadecane	Polystyrene	Fang [66]
Bromo-hexadecane	Aminoplast	Song [32]
<i>n</i> -Octadecane	Polyurea (TDI-EDA/DETA/Jeffamine T403)	Zhang [34], Cho [37], Siddhan [38], Su [39]
Butyl stearate	Polyurea (TDI-EDA)	Chen [35]
Hexadecane	Polyurea (TDI-EDA)	Zou [36]
<i>n</i> -Pentadecane/ <i>n</i> -eicosane/paraffin	UF	Tseng [40]
<i>n</i> -Hexadecane	Polystyrene	Ai [44]
Paraffin wax PRS, tetradecane, Rubitherm 20, nonadecane	Polystyrene	Sánchez [45,46]
<i>n</i> -Octadecane	Polystyrene	You [47,48], Li [49]
<i>n</i> -Octadecane	Polydivinylbenzene (PDVB)	Chaiyasat [50]
<i>n</i> -Octadecane	MMA-BDDA, MMA-DVB, MMA-TMPTA, MMA-PETRA	Qiu [51]
Paraffin, methyl stearate	Polyurethane	Cheng [52]
<i>n</i> -Docasane	PMMA	Alkan [54], Chen [61]
<i>n</i> -Eicosane	PMMA	Alkan [55]
<i>n</i> -Octadecane	PMMA	Sari [57]
<i>n</i> -Heptadecane	PMMA	Sari [56]
Paraffin	PMMA	Ma [58]
<i>n</i> -Hexadecane	PMMA	Alay [59]
Paraffin	Polystyrene	Luo [60]
<i>n</i> -Hexadecane	UF	Li [62]
Paraffin wax RT 80	Styrene-butyl acrylate (St-BA)	Fuensanta [63]
Palmitic acid (PA)	Aluminium hydroxide	Pan [42]
Paraffin	Silicon dioxide	Li [43]

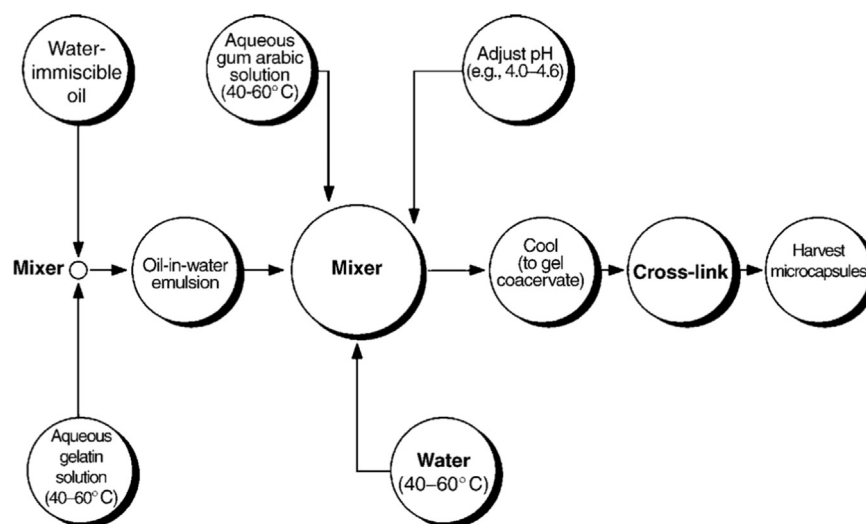


Fig. 10. Flow diagram of a typical encapsulation process based on the complex coacervation [65].

reaction temperature of 80–90 °C. The maximum measured energy storage capacity was obtained as 173.9 kJ/kg at a temperature of 51 °C. Wang et al. [81] recently encapsulated DSP with PMMA and UF. They obtained mean MEPCM diameters of 6 μm and 500 nm with corresponding latent heat capacity of 142.90 kJ/kg and 121.20 kJ/kg. However, the encapsulation efficiencies of UF and PMMA shell were only 55.74% and 63.97%, respectively, and also noticed that the organic solvent residual in MEPCM could cause the leakage problem after few multiple heating–cooling cycles. Huang et al. [82] further encapsulated a hydrated salt ($\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$) as phase change energy storage material by modifying PMMA. The morphology study revealed the

mean diameter of the capsules to be about 6.8 μm. The thermal analysis showed that the weight loss in the MEPCM was less than 10% during the heating process from 30 to 84 °C. The phase change melting point was 51 °C, whereas the heat of fusion was obtained as 150 kJ/kg.

3.5. Other micro-/nano-encapsulation methods

Apart from the already discussed encapsulation technologies, other technologies such as phase separation method, internal

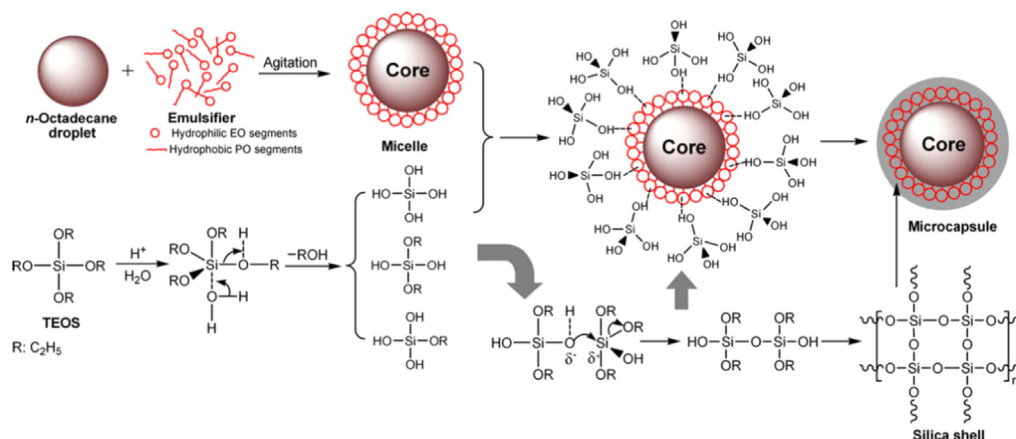


Fig. 11. Schematic formation of a sol-gel process [71].

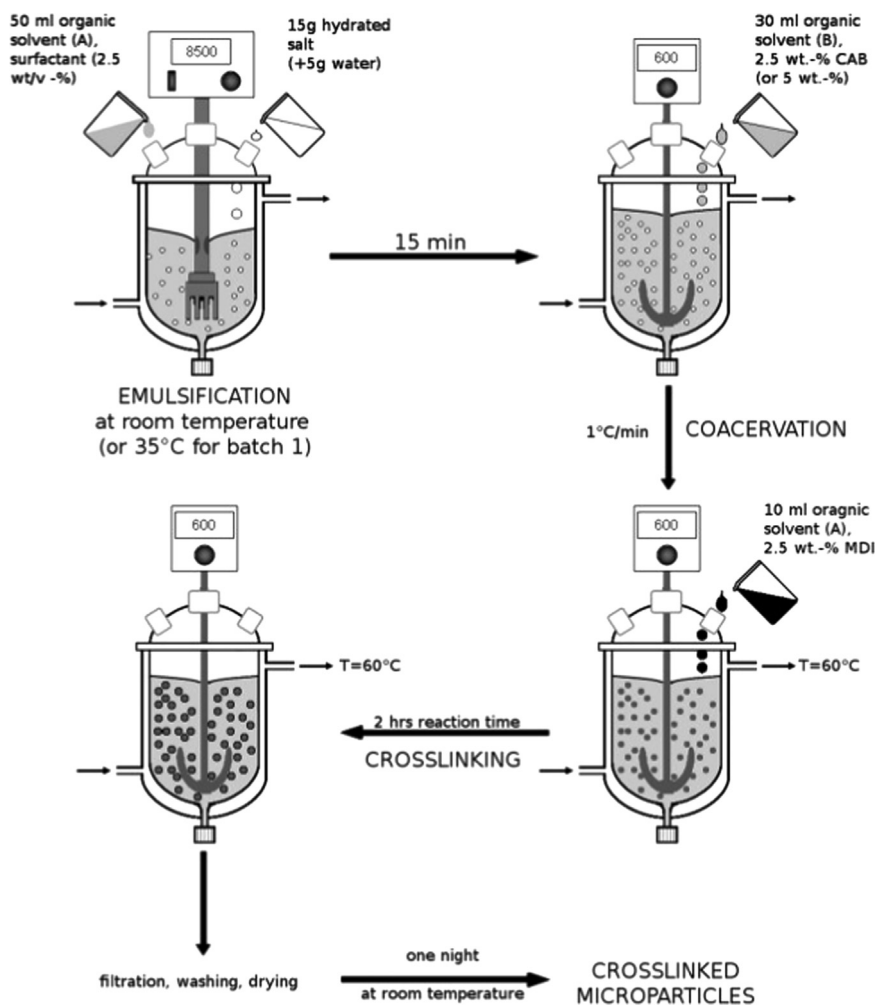


Fig. 12. Schematic overview of microencapsulation by solvent extraction/evaporation method [79].

phase separation method, pre-polymer mixing method and self-assembling method do exist.

As shown in Fig. 13, phase separation method process involves (1) preparation of shell material (PMMA) and dissolving the PMMA in the mixture of volatile solvent (v.s.), i.e., dichloromethane and non-volatile non-solvent (n.v.n.s.), i.e., hexadecane, (2) production of O/W emulsion, (3) migration of polymer-rich phase on core material particles surface by evaporation parts of v.s., and (4) removal all of v.s. to finish off encapsulation. Loxley and Vincent [83] successfully encapsulated *n*-hexadecane with polymethylmethacrylate (PMMA)

by using this method. Yang et al. [84] also used the same method to produce microencapsulated *n*-tetradecane with different shell materials (Acrylonitrile–styrene copolymer (AS), acrylonitrile–styrene–butadiene copolymer (ABS) and polycarbonate (PC)). They achieved particle sizes of less than 1 μm with melting enthalpy of more than 100 kJ/kg and encapsulation efficiency of 66–75% for all the three shell materials.

Jiang et al. [85] produced microcapsules with phenolic resin (PFR) shell and *n*-hexadecane core by internal phase separation method, which is similar to the phase separation method. The

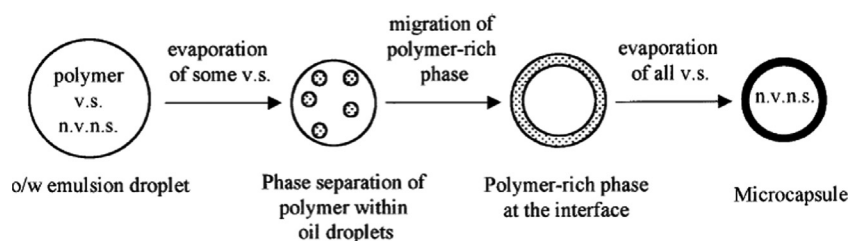


Fig. 13. Schematic of the encapsulation process [83].

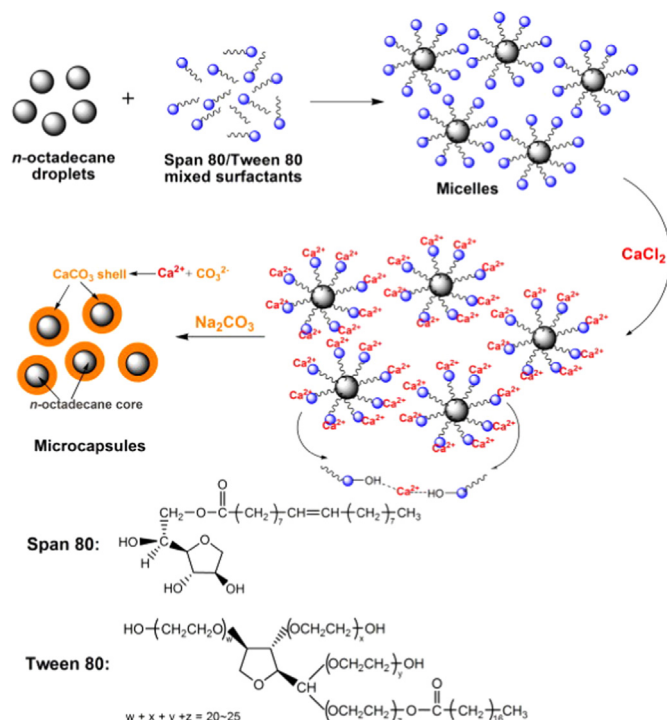
Fig. 14. Scheme of MEPCM formation with CaCO_3 shell via a self-assembly method [88].

Table 8

Statistics results of MEPCM/NEPCM via various encapsulation technologies.

Microencapsulation method	Diameter [μm]	Core content (%)	Encapsulation efficiency (%)
In-situ interfacial polycondensation	0.5–1000	29–80	71–87
In-situ suspension polymerization	0.72–237	20–75.3	78–100
In-situ emulsion polymerization	0.14–2	28–61	82.6
In-situ miniemulsion polymerization	0.1–0.5	8–60	80–82.2
Complex coacervation	2–1200	26–67	80–95
Sol-gel encapsulation	2–30	46–74	82–90.7
Phase separation method	0.5–1000	43–75	66–75
Solvent extraction/evaporation methods	0.5–10	43–53	56–64

microcapsules exhibited smooth and perfect structure but the melting point temperature did slightly reduce in comparison with pure *n*-hexadecane and also supercooling was observed upon crystallization. Zhang et al. [86] produced 1–2 μm diameter encapsulated *n*-tetradecane with PMMA and polystyrene (PS) using the same method. The results indicated that optimal microcapsules could be produced with a core/shell ratio of 3:1 and achieve a latent heat capacity of 151 kJ/kg.

The pre-polymer mixing method covers the process of dispersing core material within polymer (shell material) followed by preparation of oil/water emulsion before encapsulation. Kim and Kim [87] used this method to encapsulate *n*-octadecane with waterborne polyurethane (WBPU) shell to obtain microcapsules of sizes ranging from 1 to 6 μm. The results showed the sizes to be reducing with increasing amount of emulsifier. The heat of fusion, heat of crystallization, and encapsulation efficiencies of *n*-octadecane were also found to increase with increasing amount of microencapsulated blends, thickener and hardener.

Self-assembling method involves a process whereby a disordered system of pre-existing components form an organized structure without external direction. As shown in Fig. 14, Yu et al. [88] used this method to encapsulate *n*-octadecane with calcium carbonate (CaCO_3) as a shell material. Different core/mass ratios of 30/70, 40/60 and 50/50 were used for the fabrication to obtain corresponding energy storage capacities of 46.93 kJ/kg, 67.91 kJ/kg and 84.37 kJ/kg. Thermal conductivities ranging from 1.264 to 1.674 W/m K were obtained but more than 60% of the *n*-octadecane was wasted during the encapsulation process due to low encapsulation efficiency.

3.6. Comparison of various microencapsulation technologies

The review has shown that micro-/nano-organic PCMs can be encapsulated by various methods such as in-situ polymerization which include interfacial polycondensation, suspension polymerization and emulsion/miniemulsion polymerization, complex coacervation, and sol-gel methods, etc. In brief, in-situ suspension polymerization and complex coacervation method can be used to produce large particle size MEPCMs with high core material content and encapsulation efficiency. In-situ emulsion/miniemulsion polymerization methods are suitable for producing nanocapsules. However, in-situ suspension-like and in-situ emulsion/miniemulsion polymerization methods are hardly applied to large-scale production. This is due to the fact that high stirring rate is needed to generate oil droplets during encapsulation process which in turn results in high energy consumption and production costs. For instance, the emulsion speed used for in the reviewed in-situ polymerization encapsulation processes was found to be in the range of 1000–13,500 rpm depending on the particle size. In contrast, the polymerization reaction process was achieved with a relatively lower stirring speed of 300–600 rpm. Regarding the other processes covering sol-gel, in-situ polymerization and self-assembling methods, they can all be used to encapsulate organic PCMs using inorganic shells in order to produce MEPCM with high thermal conductivity and good fire resistance. However, inorganic shells have poor mechanical strength and are not as flexible as copolymers which could impact on the life cycle of the MEPCM.

In contrast, encapsulation of micro-/nano-inorganic PCMs is limited to only solvent extraction/evaporation method. Microencapsulated inorganic PCMs are more expensive than organic PCMs due to the large amount of non-volatile organic solvents (i.e. toluene, carbon tetrachloride, chloroform and acetone) which are

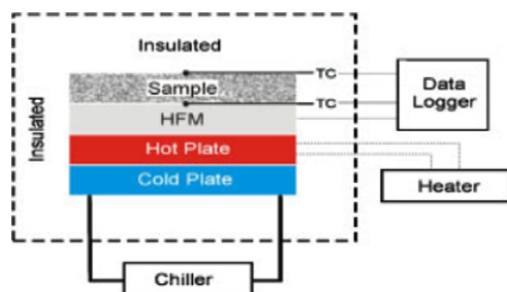


Fig. 15. Thermal conductivity testing rig.

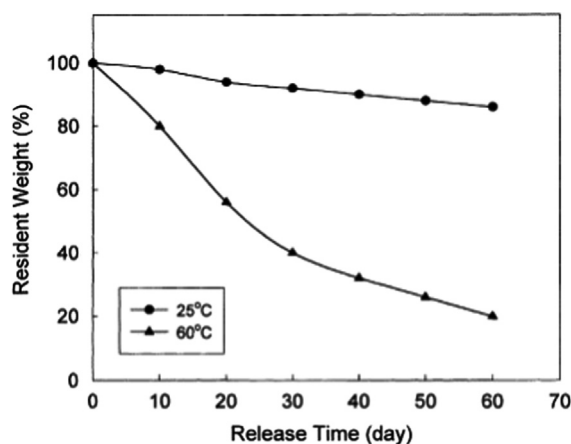


Fig. 16. Resident weight (%) of melamine resin microcapsules in long self-life test [30].

employed within the solvent extraction/evaporation encapsulation process. Beside this the residual organic solvent in MEPCM could permeate the shell materials (e.g. UF, PMMA, CAB-MDI, etc.) and cause leakage problem after a period of thermal cycling. In general the encapsulation processes have shown that the higher the emulsion speed the smaller the sizes of the capsules. Meanwhile, microencapsulation efficiencies do decrease as the ratios of core-to-coating increase. See as the summarized data in Table 8.

The review has also highlighted that the crosslinking takes the longest period in the whole microencapsulation process usually 2–5 h, but could take as long as 24 h for silicon-based shells. Apart from the crosslinking time most shell monomers require additional and external heating period to finish off copolymerization, while gelatin-gum Arabic materials need to be cooled down to about 5–10 °C. Meanwhile, all the major manufacturers of these microencapsulation shell materials have cautioned that these materials could be harmful to human and the environment. There is therefore the need for the dosage of the materials to be optimized in order to achieve complete reaction and to neutralise any unreacted reagents after crosslinking.

4. Evaluation of MEPCM/NEPCM

There are many standard instruments and methods for evaluating the properties of MEPCM/NEPCM capsules. For instance, differential scanning calorimetry (DSC) [89] can be used to measure the phase change latent enthalpy and phase change temperature; thermal gravimetric analysis (TGA) can be used the measurement of thermal stability. Thermal conductivities can be obtained with instruments such as laser thermal diffusivity-conductivity instrument (Germany, NETZSCH, LFA447) [81], Thermal Property Analyzer (USA, Decagon Devices, KD2 Pro) [90], EKO HC-110 thermal conductivity tester [71], and thermal conductivity

apparatus (Cussons Technology) [91]. Various microscopes (i.e. scanning electron microscopy (SEM), transmission electron microscope (TEM) and optical microscope) and dynamic light scattering particle size analyser could be used to study the particle size distribution and morphology of MEPCM/NEPCM. Fourier transform infrared (FT-IR) spectroscopy and X-ray diffraction (XRD) methods are also available for analysing the chemical structure of MEPCM/NEPCM shell materials. Apart from these technologies, other methods have been improved or developed for evaluating the properties of MEPCM/NEPCM.

4.1. Thermal energy storage capacity and phase change temperature

As an improvement on the traditional DSC equipment, Wang et al. [92] used an optical DSC system to record and analyse the thermal properties of different composite PCMs before and after their phase change processes. They reported high measurement accuracy of phase change temperatures and enthalpies of less than 5% in comparison with the theoretical values.

In order to determine latent heat and melting temperature of larger samples, Zhang et al. [93] developed the T-history method. Eva Günther et al. [94] compared the T-history method with two types of DSC measurement modes: dynamic mode and isothermal step mode. They found out that the isothermal step mode offers higher precision than dynamic mode, but the T-history method is more suitable for heterogeneous PCMs. Desgrosseilliers et al. [95] have since then improved on the T-history enthalpy model and been successfully used for evaluating composition-dependent two-phase equilibrium processes.

Besides evaluate MEPCM/NEPCM thermal energy storage capacity, latent heats measurement results can be used to calculate the core material content ratio in MEPCM/NEPCM by comparing the thermal energy storage capacity with pure PCMs. Meanwhile, Tseng et al. [40] assumed the MEPCM particle was a perfect sphere and then through core material content and MEPCM diameter calculated the shell thickness. Moreover, it is helpful for evaluation of micro-/nano-encapsulation process, i.e., Zhang et al. [34] suggested using encapsulation efficiency, which was defined as the ratio of the actual core content of the microcapsules to the theoretical core content.

4.2. Thermal conductivity

For the evaluation of thermal conductivity of NEPCM/MEPCM, researchers such as Meng and Wang [96], Wang et al. [97] and Hunger et al. [98] have built and tested other devices using the hot wire method. Meanwhile, the effective thermal conductivities of MEPCMs and their compositions have been studied with various methods. Zhao et al. [99] used the 3ω method [100] in an experimental study and achieved the maximum effective thermal conductivity of the MEPCM at its peak value of the phase change temperature. They also observed that the conductivity value did increase as the density of the material was increased. This was attributed to reduction in the porosity of the MEPCM. Darkwa and Kim [101] developed an effective thermal conductivity testing rig in accordance with ISO 8301 Standards and used it to determine the thermal conductivity of a composite MEPCM sample as shown in Fig. 15. Similarly, Marchi et al. [102] tested the thermal conductivity of panels containing MEPCM by using the thermofluximeter method and in accordance with the EN12667:2001 Standards. The experimental results were closely in agreement with the theoretical predictions based on the effective medium theory (EMT) equations.

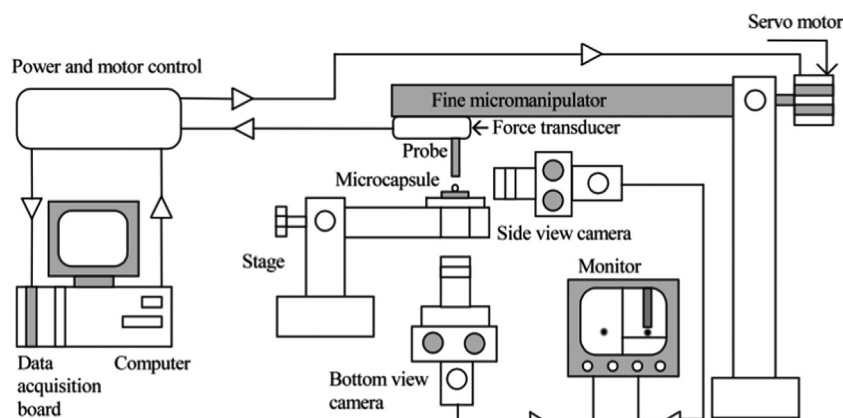


Fig. 17. Schematic diagram of the micromanipulation rig [105].

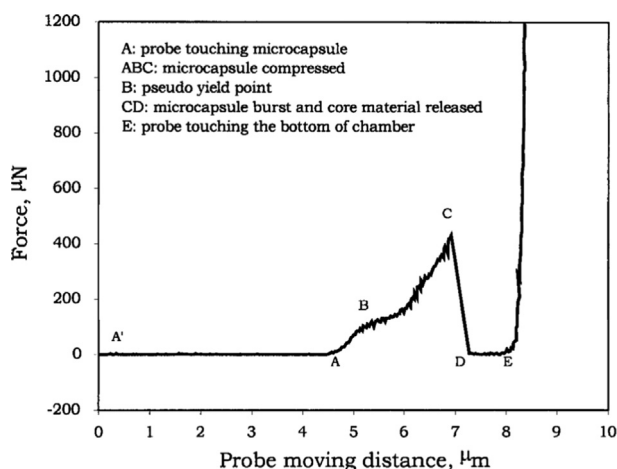


Fig. 18. Force versus probe moving distance [105].

4.3. Thermal stability

Thermal stability of MEPCMs is an important property to ensure long term usage. Hong and Park [30] investigated the long-term stability of MEPCM under a 60-day drying experimental test. As shown in Fig. 16 a known weight sample of MEPCM was dried under 25 °C and 60 °C, and weighed after every 10 days for the resident weight percentage. The results show that the progressive residual weight percentage at 60 °C drying condition was about 42%, whereas under 25 °C it was at the rate of 8% after 60 days. Sharma et al. [103] proposed an accelerated thermal cycle testing method to study the changes in latent heat and melting temperature of acetamide, stearic acid and paraffin wax. The results showed good thermal stability for the paraffin and acetamide after 1500 thermal cycles. Alkan et al. [54] also performed a series of thermal stability tests on PMMA/docosane microcapsules using their DSC curves and FT-IR data and found no significant changes in their phase change temperatures, latent heat enthalpies and chemical structures after a number of thermal cycles.

4.4. Mechanical strength

Zhang et al. [104], Sun and Zhang [105,106] and Hu et al. [107] have all through the micromanipulation technology (shown in Fig. 17) determined the mechanical properties of various MEPCMs/NEPCMs. It involves using a single probe in squeezing a microcapsule, which is positioned on a slide, and then measuring its bursting strength. The probe is connected to a force transducer and mounted on a 3D micromanipulator that can be programmed to travel at a given speed. The slide is placed on the stage of an

inverted microscope. When a microcapsule is squeezed, the force being imposed on it is measured simultaneously by sampling the voltage signal from the force transducer. When the probe eventually touches the slide, the force will increase rapidly to signal the end of the test as displayed in Fig. 18. Sun and Zhang's investigation [105] showed that the mean bursting force and yield point did not change significantly with different compression speed.

Su et al. [108] studied the mechanical properties of microcapsules covered with M/F shells by using their SEM photographs and different compression loads. They concluded that when the mass ratio of the core and shell material was 3:1, a yield point was achieved with a compression load of about 1.1×10^5 Pa. They also reported that the microcapsules displayed a form of plastic behaviour when the compression load was increased beyond that point. The double shells were however found to possess better mechanical properties. Darkwa et al. [90] also investigated mechanical strength of bulk of microcapsules with similar method and found no sign of fragmentation of the MEPCM particles after a pressure of 2.8 MPa was applied.

4.5. Chemical stability

Anti-osmosis test is an important approach for determining the rate of weight loss and thus chemical stability of MEPCMs. Zhang et al. [34] used this method for evaluating the durability of samples of MEPCMs. The releasing rate (weight percentage of the released substance) of 10 g of microencapsulated *n*-octadecane was measured by dispersing it in a 50 ml acetone (as an extraction solvent) and at a stirring speed of 200 rpm. Analysis of the results showed that the release rate of the microcapsules increased when the weight percentage of the core materials increased and that the shell produced with Jeffamine material exhibited the best anti-osmosis behaviour. Zhang et al. [71] capitalised on this finding and evaluated SiO₂ coated *n*-octadecane with the same method. Su et al. [39] also used this method but with a different extraction solvent (ethyl alcohol) to evaluate another type of MEPCM. They however found out that the amount of emulsifier used in that investigation had some effect on the material's anti-osmosis property.

5. Conclusions

This paper was focused on the review of various types of solid-liquid PCMs (i.e., organic, inorganic and eutectics) and their thermophysical properties for various energy storage applications. The review has shown that organic solid-liquid PCMs have more advantages in terms of wider temperature range application, less

super cooling and segregation effects, and other capabilities. They do however have disadvantages such as low thermal conductivity, low density and being flammable. In contrast, inorganic solid-liquid PCMs possess higher heat storage capacities and conductivities, cheaper and readily available as well as being non-flammable. Nevertheless inorganic PCMs do experience problems associated with supercooling and phase segregation during the phase change process. Concerning eutectic PCMs, they were found to have an advantage over other types of PCMs in the sense that their melting points can be adjusted by combining different weight percentages of components. In addition, they possess high thermal conductivity and density, but have low latent and specific heat capacities.

The paper also examined the technologies currently being employed for encapsulation of PCMs and the limitations of some of encapsulation shell materials. For instance some inorganic shells are good for encapsulation and thermal conductivity enhancement of PCMs but do experience low encapsulation efficiency, longer crosslinking time and low mechanical strength. It was however identified that thermal and chemical stability and the mechanical strength of encapsulated PCMs could strongly be influenced by the morphology of their particles. Further development towards the enhancement of these properties is therefore encouraged. In considering all these factors it could be deduced that in-situ polymerization method offers the best technological approach in terms of encapsulation efficiency and structural integrity of core material. Regarding evaluation methods, there are number of standalone systems which have proved to be reliable but may require some form of standardisation especially in the area of evaluation of mechanical properties.

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